

EXERCISE

- Solve the following differential equations using the method of variation of parameter.

1. $y'' + y = \cos x,$

2. $y'' + y = \sin x,$

3. $y'' + 4y = \tan 2x,$

4. $y'' + y = e^x,$

5. $y'' + a^2y = \sec ax,$

6. $y'' + a^2y = \cot ax,$

7. $y'' - 2y' = e^x \cos x,$

8. $y''' + y' = \operatorname{cosec} x,$

9. $y'' + 2y' + y = \frac{1}{x^2 e^x},$

10. $y'' + 4y = \operatorname{cosec} 2x,$

11. $y'' - 6y' + 9y = \frac{e^{3x}}{x^2},$

12. $y'' - 4y' + 4y = e^{2x},$

13. $y'' - y' - 2y = x^2$

14. $y'' + 4y' + 5y = e^{-2x} \sec x,$

15. $y'' - 2y' + y = e^x \sin x,$

16. $y'' + 2y' + y = e^{-x} \cos x,$

17. $y'' - y = \frac{2}{1 + e^x},$

18. $y'' - 3y' + 2y = \frac{e^x}{1 + e^x},$

19. $y'' - 4y' + 3y = \frac{e^x}{1 + e^x},$

20. $y'' - 2y' + y = xe^x \log x,$

21. $y'' - 2y' + 2y = e^x \tan x,$

22. $y'' + y = \frac{1}{1 + \sin x},$

23. $y'' + 3y' + 2y = 2e^x.$

24. $y'' - y = xe^x,$

25. $y'' - 2y' = e^x \sin x,$

26. $y'' - 2y' + y = xe^x \sin x,$

27. Solve $(2x + 1)(x + 1)y_2 + 2xy_1 - 2y = (2x + 1)^2$ by the method of variation of parameter, given that $y = x, y = \frac{1}{x+1}$ are two linearly independent solutions of the corresponding homogeneous equation.

28. Solve $x^2y_2 - x(x + 2)y_1 + (x + 2)y = x^3$ by the method of variation of parameter, given that $y = x, y = xe^x$ are two linearly independent solutions of the corresponding homogeneous equation.

29. Solve $x^2y_2 - xy_1 - 3y = x^2 \log x$ by the method of variation of parameter, given that $y = x^3, y = \frac{1}{x}$ are two linearly independent solutions of the corresponding homogeneous equation.

30. Solve $x^2y_2 + xy_1 - y = x^2e^x$ by the method of variation of parameter, given that $y = x, y = \frac{1}{x}$ are two linearly independent solutions of the corresponding homogeneous equation.

31. Solve $x^2y_2 - xy_1 = x^3e^x$ by the method of variation of parameter, given that $y = x^2, y = 1$ are two linearly independent solutions of the corresponding homogeneous equation.

32. Solve $x^2 y_2 - 2x(x+1)y_1 + 2(1+x)y = x^3$ by method of variation of parameters, given that $y = x$, $y = xe^{2x}$ are two linearly independent solutions of the corresponding homogeneous equation.
33. Solve $x^3 y_3 + 3x^2 y_2 = 1$ by method of variation of parameter, given that $y = 1$, $y = x$, $y = \frac{1}{x}$ are three linearly independent solutions of the corresponding homogeneous equation.

ANSWERS

1. $y = c_1 \cos x + c_2 \sin x + \frac{1}{2} x \sin x$, 2. $y = c_1 \cos x + c_2 \sin x - \frac{1}{2} x \cos x$,
3. $y = c_1 \cos 2x + c_2 \sin 2x - \frac{1}{4} \cos 2x \ln |\sec 2x + \tan 2x|$
4. $y = c_1 \cos x + c_2 \sin x + \frac{1}{2} e^x$,
5. $y = c_1 \cos ax + c_2 \sin ax - \frac{1}{a^2} \cos ax \ln |\sec ax| + \frac{x}{a} \sin ax$,
6. $y = c_1 \cos ax + c_2 \sin ax + \frac{1}{a^2} \sin ax \ln \tan \left(\frac{ax}{2} \right)$,
7. $y = c_1 + c_2 e^{2x} - \frac{1}{2} e^x \cos x$,
8. $y = c_1 \cos x + c_2 \sin x + c_3 - \cos x \ln |\sin x| - x \sin x + \ln |\operatorname{cosec} x - \cot x|$
9. $y = e^{-x} [c_1 x + c_2 - \ln x]$
10. $y = c_1 \cos 2x + c_2 \sin 2x + \frac{1}{4} \sin 2x \ln |\sin 2x| - \frac{x}{2} \cos 2x$,
11. $y = e^{3x} [c_1 x + c_2 - \ln x]$ 12. $y = e^{2x} \left[c_1 x + c_2 + \frac{x^2}{2} \right]$
13. $y = c_1 e^{-x} + c_2 e^{2x} - \frac{x^2}{2} + \frac{x}{2} - \frac{3}{4}$,
14. $y = e^{-2x} (c_1 \cos x + c_2 \sin x) + x e^{-2x} \sin x + e^{-2x} \cos x \ln (\cos x)$,
15. $y = (c_1 + c_2 x - \sin x) e^x$, 16. $y = (c_1 + c_2 x - \cos x) e^{-x}$,
17. $y = a e^x + b e^{-x} - 1 - x e^x + (e^x - e^{-x}) \ln (1 + e^x)$,
18. $y = a e^x + b e^{2x} - x e^x - x e^{2x} + e^x \log (e^x + 1) + e^{2x} \log (e^x + 1)$,
19. $y = a e^x + b e^{3x} + \frac{1}{2} e^{2x} + \frac{1}{2} (1 - e^{2x}) e^x \log (1 + e^{-x})$,
20. $y = (a + bx) e^x + \frac{1}{6} x^3 e^x \log x - \frac{5}{36} x^3 e^x$,

21. $y = e^x [a \cos x + b \sin x - \cos x]$
22. $y = a \cos x + b \sin x - 1 + \sin x$
23. $y = \frac{a}{x} + \frac{b}{x^2} + \frac{1}{3} e^x$,
25. $y = a + b e^{2x} - \frac{1}{2} e^x \sin x$
27. $y = ax + \frac{b}{x+1} + \frac{x^2}{6} \frac{4x}{x}$
29. $y = ax^3 + \frac{b}{x} - \frac{1}{3} x^2 (\log x)$
31. $y = a + bx^2 + (x-1)$
33. $y = a + bx + \frac{c}{x} - \log x$

6.7 Method of U

The method of undetermined coefficients for solving a differential equation of variation of parameters is discussed later on. V

$F(D)y$

where $D = \frac{d}{dx}$.

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where $q_1(x)$, q_2 differentiation

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of variation of parameter,
nt solutions of the corre-
meter, given that $y = 1$,
he corresponding homo-

$$\sin x - \frac{1}{2}x \cos x,$$

$$\sec x - \cot x |$$

$$\left[\begin{array}{c} \text{the constant} \\ \text{the value of } x \\ \text{the value of } y \end{array} \right]$$

$$21. y = e^x [a \cos x + b \sin x - \cos x \log(\sin x + \tan x)],$$

$$22. y = a \cos x + b \sin x - 1 + \sin x - x \cos x + \sin x \log(1 + \sin x),$$

$$23. y = \frac{a}{x} + \frac{b}{x^2} + \frac{1}{3}e^x,$$

$$24. y = Ae^x + Be^{-x} + \frac{x(x-1)e^x}{4},$$

$$25. y = a + be^{2x} - \frac{1}{2}e^x \sin x,$$

$$26. y = ae^x + bxe^x - xe^x \sin x - 2e^x \cos x,$$

$$27. y = ax + \frac{b}{x+1} + \frac{x^2}{6} \frac{4x+3}{x+1},$$

$$28. y = ax + bxe^x - x(x+1),$$

$$29. y = ax^3 + \frac{b}{x} - \frac{1}{3}x^2 \left(\log x + \frac{2}{3} \right),$$

$$30. y = ax + \frac{b}{x} + e^x \left(1 - \frac{1}{x} \right),$$

$$31. y = a + bx^2 + (x-1)e^x,$$

$$32. y = ax + bxe^{2x} - \frac{x}{4} - \frac{x^2}{2},$$

$$33. y = a + bx + \frac{c}{x} - \log x.$$

6.7 Method of Undetermined Coefficients

The method of undetermined coefficients is also applied to find a particular solution of a differential equation. Though this method is simpler and economical than the method of variation of parameter, this method cannot be applied in a few cases which are discussed later on. We consider the problem of solving an equation of the form

$$F(D)y = [D^n + P_1 D^{n-1} + P_2 D^{n-2} + \dots + P_{n-1} D + P_n]y = Q, \quad (i)$$

$$\text{where } D = \frac{d}{dx}.$$

In this method we assume the particular integral as

$$y_p = A_1 q_1(x) + A_2 q_2(x) + \dots + A_s q_s(x), \quad (ii)$$

where $q_1(x), q_2(x), \dots, q_s(x)$ are the terms of Q and those arising from these terms by differentiation, for example, if the equation is $F(D)y = \sin x$, we take for (ii),

$$y_p = A \cos x + B \sin x,$$

if the equation is $F(D)y = x^2$, we take for (ii),

$$y_p = Ax^2 + Bx + C,$$

if the equation is $F(D)y = e^{2x} + e^{4x}$, we take for (ii),

$$y_p = Ae^{2x} + Be^{4x},$$

where A 's are undetermined

If a term of Q , say L , is also a term of the complementary function corresponding to an s -fold root m , then in (ii) we add a term $x^s L$ and the terms arising from it by differentiation. For example, for the equation $(D-1)^2(D+2)y = e^x + \sin x$, we take the particular integral as

$$y_p = Ax^2e^x + Bxe^x + Ce^x + D\sin x + E\cos x.$$

If a term of Q is $x^s L$ and L is also a term of the complementary function corresponding to an s -fold root m , then in (ii) we add a term $x^{s+s}L$ and the terms arising from it by differentiation. For example, for the equation $(D-1)^2(D+2)y = xe^x + x^3$, we take the particular integral as

$$y_p = Ax^3e^x + Bx^2e^x + Cxe^x + De^x + Ex^3 + Fx^2 + Gx + H.$$

When Q is an algebraic function then the particular integral of (6.7.1) will be of the form of (ii), if all P_i are non-zero. If $P_n = 0$, then we take the particular integral as

$$y_p = x[A_1q_1(x) + A_2q_2(x) + \dots + A_sq_s(x)].$$

If $P_n = 0$ and $P_{n-1} = 0$, then we take the particular integral as

$$y_p = x^2[A_1q_1(x) + A_2q_2(x) + \dots + A_sq_s(x)] \text{ and so on.}$$

WORKING RULE

We know that the general solution of the differential equation

$$y_2 + P(x)y_1 + Q(x)y = R(x)$$

has the form

$$y = \text{C.F.} + \text{P.I.}$$

The method of undetermined coefficients leads the way of finding the P.I. in the following ways.

// **R1.** If $R(x) = e^{\lambda x}$ and λ is a multiple root of the auxiliary equation of order k , then we take the P.I. as

$$y_p = Ax^k e^{\lambda x},$$

where A is the undetermined constant. Obviously, $k = 0$, if λ is not a root of the auxiliary equation.

// **R2.** If $R(x) = a_0 + a_1x + a_2x^2 + \dots + a_kx^k$, i.e., a polynomial in x and $P(x)$, $Q(x)$ do not vanish, we take the P.I. as

$$y_p = A_0 + A_1x + A_2x^2 + \dots + A_kx^k.$$

When $Q(x) = 0$, we take

$$y_p = x(A_0 + A_1x + A_2x^2 + \dots + A_kx^k)$$

and if $P(x) = 0 = Q(x)$, then we take

$$y_p = x^2(A_0 + A_1x + A_2x^2 + \dots + A_kx^k)$$

// **R3.** If $R(x) = \sin ax$ or $\cos ax$ then we take

$$y_p = A \sin ax + B \cos ax,$$

if $\sin ax$ or $\cos ax$ is absent in the complementary function.

complementary function corresponding to the terms arising from it by the method of undetermined coefficients. If $(D-1)^2(D+2)y = e^x + \sin x$, we take $y = E \cos x$.

complementary function corresponding to the terms arising from it by the method of undetermined coefficients. If $(D-1)^2(D+2)y = xe^x + x^2$, we take $y = Fx^2 + Gx + H$.

integral of (6.7.1) will be of the form $x^r e^{ax} \sin bx$ or $x^r e^{ax} \cos bx$. The particular integral as $y_p(x)$.

integral as

] and so on.

equation

of finding the P.I. in the

equation of order k , then

λ is not a root of the

in x and $P(x), Q(x)$ do

If $\sin ax$ or $\cos ax$ is present in the complementary function then we take

$$y_p = x(A \sin ax + B \cos ax).$$

R4. If $R(x) = e^{\lambda x} \sin bx$ or $e^{\lambda x} (a_0 + a_1 x + a_2 x^2 + \dots + a_k x^k)$ or $\sin bx (a_0 + a_1 x + a_2 x^2 + \dots + a_k x^k)$, then the P.I. will be modified according to the above mentioned rules.

For example, if $R(x) = e^{\lambda x} \sin ax$ and λ is not a root of the auxiliary equation, then we take the P.I. as

$$R(x) = e^{\lambda x} (A \sin ax + B \cos ax).$$

If $R(x) = e^{\lambda x} (a_0 + a_1 x + \dots + a_k x^k)$ and λ is a multiple root of the auxiliary equation of order k , then we take

$$y_p = x^k e^{\lambda x} (A_0 + A_1 x + \dots + A_k x^k).$$

If $R(x) = \sin bx (a_0 + a_1 x + \dots + a_k x^k)$ and $\sin bx$ is present in the complementary function, then we take

$$y_p = x(A \cos bx + B \sin bx) (A_0 + A_1 x + \dots + A_k x^k).$$

WORKED-OUT EXAMPLE

1. Solve the following equation by the method of undetermined coefficients:

$$y_2 + 2y_1 + 2y = e^{2x}.$$

Solution

Let $y = e^{mx}$ be a trial solution, then the auxiliary equation is

$$m^2 + 2m + 2 = 0.$$

Solving, we get $m = -1 \pm i$. So the C.F. is

$$e^{-x} (A \cos x + B \sin x).$$

Since 2 is not a root of the auxiliary equation, we take the P.I. as

$$y_p = C e^{2x}.$$

Using this in the given equation, we get $C = \frac{1}{10}$. Hence the general solution of the given equation is

$$y = e^{-x} (A \cos x + B \sin x) + \frac{e^{2x}}{10}.$$

2. Solve the following equation by the method of undetermined coefficients:

$$y_2 + 4y_1 + 4y = 16e^{-2x}.$$

Solution

Let $y = e^{mx}$ be a trial solution, then the auxiliary equation is

$$m^2 + 4m + 4 = 0.$$

Solving, we get $m = -2, -2$. So the C.F. is

$$(c_1 + c_2 x) e^{-2x}.$$